

SOLOMON AHEDOR

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[GitHub](#)

PROFESSIONAL PROFILE

Final-year Metallurgical Engineering student at KNUST with hands-on industrial mineral processing experience from AngloGold Ashanti (Obuasi Mine) across refractory gold BIOX-CIL-AARL elution circuits, met-lab analytical instrumentation (LECO, AAS) and plant-floor flotation diagnostics. Proven track record in applied machine learning and computational materials science, authoring research manuscripts on flotation process optimization ($R^2 = 0.9935$) and ML-accelerated high-entropy alloy discovery. Seeking to leverage metallurgical engineering fundamentals, data-driven optimization, and laboratory rigor to drive process excellence.

EDUCATION

Kwame Nkrumah University of Science and Technology (KNUST) | Kumasi, Ghana Expected: Sept 2026
Bachelor of Science (BSc) in Metallurgical Engineering

- **Relevant Coursework:** Mineral Processing, Extractive Metallurgy, Physical Metallurgy, Materials Characterization (XRD, SEM, Optical), Materials Informatics & Computational Materials, Machine Learning, Chemical Thermodynamics, Process Control, Project Management.

INDUSTRIAL EXPERIENCE

AngloGold Ashanti (Obuasi Mine) | Obuasi, Ghana

Nov 2025 – Dec 2025

Metallurgy & Process Engineering Intern

- **Refractory Circuit & BIOX Monitoring:** Rotated across the full refractory gold processing plant; monitored BIOX bacterial oxidation activity and performed ferrous/ferric ($\text{Fe}^{2+}/\text{Fe}^{3+}$) ion titration determinations to evaluate sulfide oxidation rates and optimize downstream CIL adsorption kinetics.
- **AARL Elution & Desorption:** Tracked carbon-in-leach adsorption stages and Anglo-American Research Laboratories (AARL) elution and desorption circuits, monitoring loaded-carbon strip efficiency and electrowinning parameters ahead of gold smelting.
- **Met-Lab Flotation & Grind Kinetics:** Conducted laboratory batch-froth flotation tests and Particle Size Analysis (PSA) in the Metallurgical Laboratory to evaluate recovery efficiency, grind-recovery kinetics, and reagent dosing schemes for continuous circuit setting refinement.
- **Analytical Instrumentation (LECO / AAS):** Operated LECO (S832 DR & S632) combustion instrumentation for total carbon, sulfur, and carbonate determination; utilized Atomic Absorption Spectroscopy (AAS) for precision detection of aqueous gold (Au) and arsenic (As) concentrations.

SELECTED RESEARCH & ENGINEERING PROJECTS

Machine Learning Pipeline for Froth Flotation Optimization | Lead Researcher

2025 – 2026

- **Pipeline Architecture:** Formulated a dual-circuit machine learning pipeline evaluating Leave-One-Out Cross-Validation (LOOCV) on UG2 platinum trials and time-series forecasting on a 737,453-row industrial iron ore process historian ([GitHub](#)).
- **Data Leakage Elimination:** Designed an assay-event extraction algorithm eliminating 94% forward-fill synthetic data leakage; engineered interval window statistics and Autoregressive (AR) lag features, achieving $R^2 = 0.9935$ with Hybrid XGBoost (a $\Delta R^2 = +0.7003$ improvement over baseline).
- **Empirical Findings & Publication:** Established that sample-to-feature ratio $n/p \geq 5$ is required for generalizable signal, proving regularized Ridge ($R^2 = 0.954$) for linear stoichiometry and XGBoost ($R^2 = 0.741$) for non-linear cell hydraulics. Full manuscript prepared for submission to Minerals Engineering.

MetaForge: ML-Driven High-Entropy Alloy (HEA) Discovery | KNUST (Supervised by Prof. K. Mensah-Darkwa) 2025 – 2026

- **Inverse Design Platform:** Built an end-to-end computational materials discovery pipeline combining 132 Matminer Magpie descriptors, Random Forest regression, and custom Genetic Algorithm inverse design to discover lightweight, refractory, Aerospace and corrosion-resistant HEA compositions.
- **GNN Structural Validation:** Validated structural stability by relaxing 54-atom BCC and 48-atom FCC supercells with CHGNet Graph Neural Network (GNN) potentials, confirming physical lattice contraction margins (+5.1% in refractory HEAs). Deployed interactive web app on Flask/React ([GitHub](#))

Biomass-Derived Carbon Filter for Turbid Water Treatment | KNUST (Supervised by Prof. E. Gikunoo) 2025 – 2026

- **Adsorption Optimization:** Designed and fabricated a sustainable water filtration system utilizing agricultural waste (coconut shell activated carbon); optimized turbidity adsorption parameters using Response Surface Methodology (RSM) with Face-Centered Central Composite Design (CCD) in Design Expert achieving 88% turbidity removal.

TECHNICAL & ENGINEERING SKILLS

- **Metallurgical & Process Engineering:** Refractory Gold Processing, Bacterial Oxidation (BIOX), Carbon-in-Leach (CIL), AARL Elution & Electrowinning, Froth Flotation Kinetics, Grinding & Comminution, Particle Size Analysis (PSA), Response Surface Methodology (RSM / CCD).
- **Analytical & Lab Instrumentation:** LECO Carbon/Sulfur Analyzers (S832 DR, S632), Atomic Absorption Spectroscopy (AAS), XRF Soft Sensing, Optical Microscopy, Sieve Shakers.
- **Computational Materials & Machine Learning:** Python (PyTorch, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn), Materials Informatics (Pymatgen, Matminer, CHGNet GNNs, SQS Supercells), Genetic Algorithms, Time-Series Autoregressive Modeling, Model Predictive Control (MPC).
- **Engineering Software & Tools:** Design Expert, Ansys Granta Selector, MATLAB, Git / GitHub, Next.js, TypeScript, Flask, Microsoft Office Suite.

LEADERSHIP, AWARDS & CERTIFICATIONS

- **Executive Head & R&D Lead | Materials Innovation Hub (MIH), KNUST (2025–2026):**
Led the organization and oversaw student researchers across active biopolymer R&D projects, developing biodegradable materials from local agricultural waste.
- **Fellow | Aspire Leaders Program (2026):** Selected for competitive international leadership and professional development program founded by Harvard University faculty.
- **Awardee | Hult Prize Global Challenge (2025):** National Citation Recipient & 2nd Runner-Up (KNUST Campus Edition) for scalable social enterprise and technological innovation.
- **Zonal Grant Winner | Mastercard-Kosmos Innovation Center (KIC) Competition (2024–2025):**
Awarded competitive seed funding for market research and technological feasibility validation in agribusiness & material processing.

LANGUAGES

English (Fluent) | Twi (Native) | German (Basic)